ME-446 AI & INTERNET OF THINGS | COMPLEX ENGINEERING PROBLEM | FALL 2026

AI-Based Building Energy Consumption Prediction and Analysis for HVAC and Energy Efficiency Applications

NED University of Engineering and Technology | Department of Mechanical Engineering
Group: [Group No.] **Students:** [Name — Roll No.] **Instructor:** [Instructor Name] **Submission:** 4 November 2026

1. Engineering problem

Buildings and HVAC systems account for a major share of electricity use. A model of normal hourly demand can support monitoring, demand planning, and anomaly screening. The engineering question is not only "how accurate is the model?" but "accurate for which buildings, and under what conditions?"

Target: hourly electricity *meter_reading* for meter 0.
Inputs: building metadata, calendar, weather, building identity.
Unit caveat: dataset energy units (documented as kWh), not independently verified; not power, efficiency, or savings.

2. Objectives

- Audit the ASHRAE dataset and document data-quality decisions.
- Build a leakage-safe preprocessing and feature pipeline.
- Compare Linear Regression, Random Forest, and HistGradientBoosting.
- Select on validation data; report test metrics separately.
- Quantify unseen-building generalization and demonstrate two unseen cases.
- Produce engineering insights, Tableau-ready data, and a build guide.

3. Dataset and scope

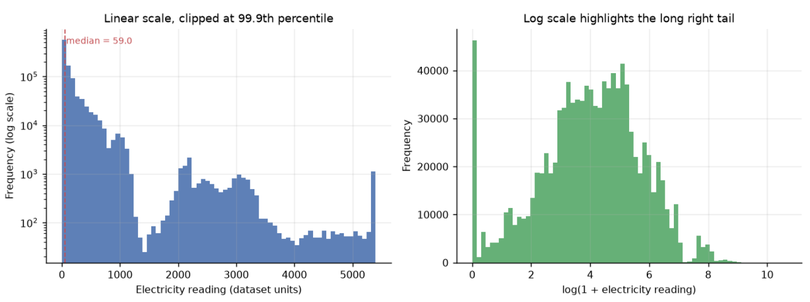
Item	Value
Full train rows	20,216,100
Electricity rows	12,060,910
Electricity buildings	1,413
Local subset	120 buildings / 1,030,967 rows
Sites / primary uses	16 / 16
Train / validation / test	769k / 175k / 87k

All reported metrics are subset metrics. The subset preserves all 16 sites and all 16 uses, but not the full building mix. Dataset approval and uniqueness are pending.

4. Data quality and features

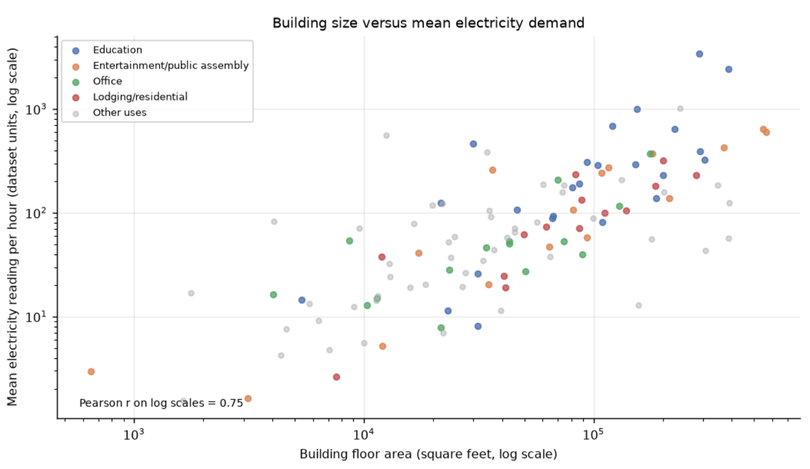
- No duplicate building–meter–timestamp keys in electricity data.
- 4.40% zero electricity readings retained; zeros are part of normal operation.
- Missing metadata retained: *year_built* 51.76%, *floor_count* 70.90% in subset.
- Weather: past-only forward fill (3 h), negative precipitation treated as missing, pressure 800–1100 hPa, missing flags.
- **24 features:** building/site identity, primary use, log floor area, year built, floor count, calendar, cyclical hour/month, temperature, dew point, cloud, precipitation, pressure, missing flags.
- No target lags, no future weather fill, no test-driven feature selection.

Demand is right-skewed



Median 59.05 vs mean 188.78; p99 2,454; heavy right tail.

Size is a strong, interpretable signal



Log floor area vs log reading: $r = 0.750$; useful for segmented benchmarking.

5. Methodology

Pipeline: raw CSVs → typed Parquet → electricity filter → metadata/weather joins → feature engineering → chronological split → pipelines → validation → test → error/generalization analysis → dashboard extracts.

Chronological split: train Jan–Sep 2016; validation Oct–Nov; test Dec. No random shuffling across time.

6. Models and selection

Model	Key configuration
Linear Regression	One-hot; median impute; scale
Random Forest	80 trees; depth 16; 250k sample
HistGradientBoosting	200 iters; lr 0.08; 31 leaves

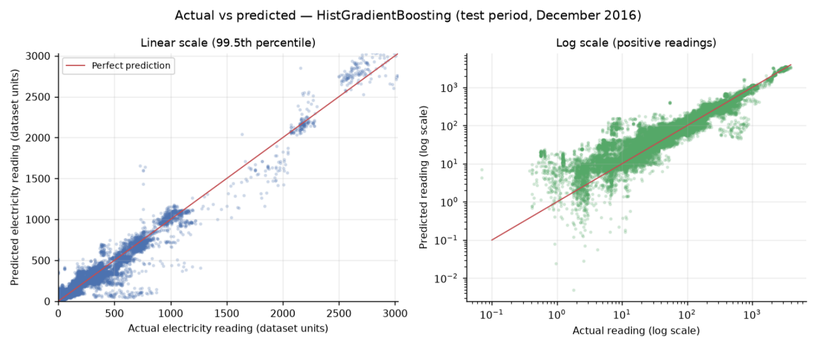
Selection rule: **lowest validation RMSE**. Test evaluated once after selection. Seed 42. Metrics: MAE, RMSE, R^2, RMSLE.

7. Results

Model	Test MAE	Test RMSE	Test R^2	RMSLE
Mean baseline	209.57	406.05	−0.001	1.920
Linear Regression	61.39	154.51	0.855	0.942
Random Forest	**32.05**	78.90	0.962	**0.448**
HistGradientBoosting	33.20	**69.78**	**0.970**	0.591

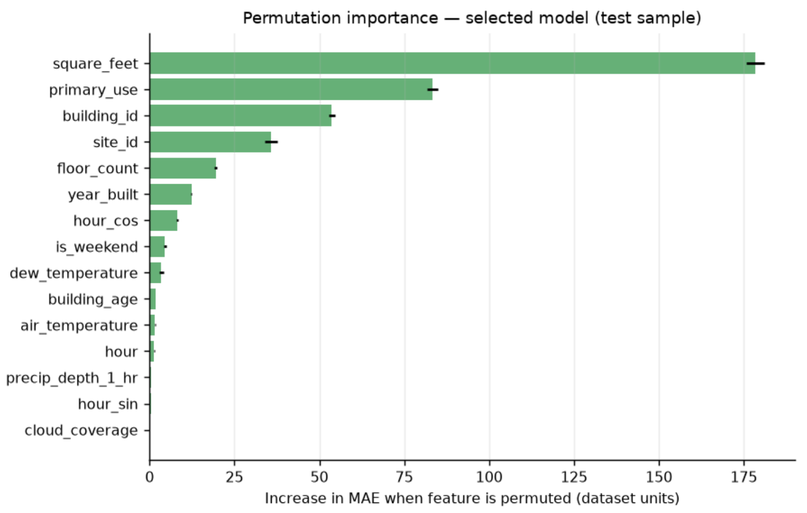
Selected: HistGradientBoosting, by validation RMSE (165.74 vs 166.92). Random Forest has lower MAE and RMSLE; the trade-off is reported. HGB fit about 4.3× faster.

Diagnostics: errors concentrate at high demand



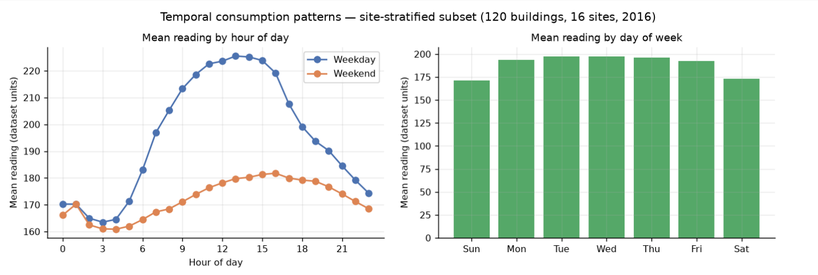
Test actual vs predicted; the 45° line is a perfect forecast. High-demand points are harder.

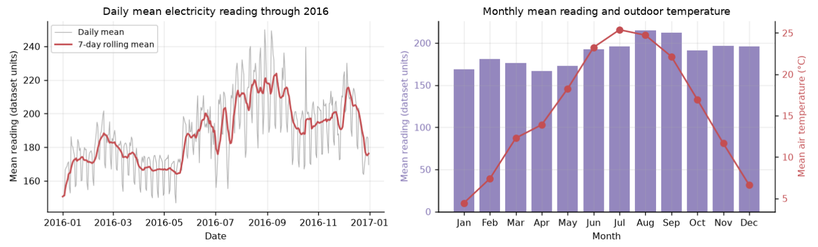
What drives the model



Building identity dominates; weather and hour add useful signal.

Occupancy creates a weekday/midday cycle





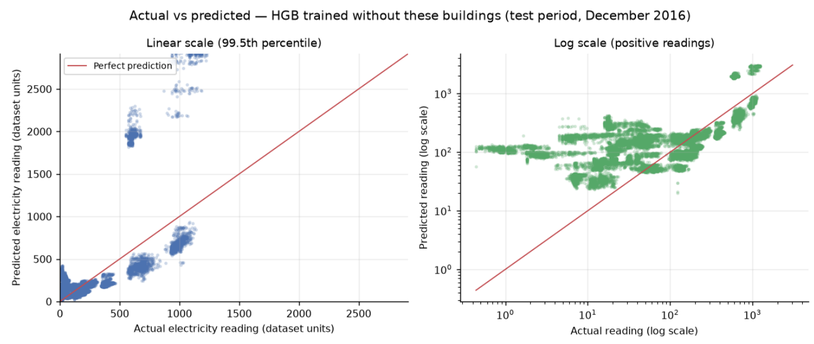
Weekday mean 195.44 vs weekend 172.24; peak 13:00, trough 03:00.

8. Critical finding: unseen buildings

Group	MAE	RMSE	R^2	RMSLE
Seen buildings, test	33.86	74.15	0.971	0.595
Unseen buildings, test	150.51	321.33	**−0.555**	1.525

High seen-building R^2 is building-specific interpolation, not a universal physical model. The model relies on *building_id*. Unseen predictions are biased high (242.49 vs 176.99). Do not deploy to new buildings without calibration or richer descriptors.

Unseen building holdout



Building holdout: predictions compress toward the mean, giving negative R^2.

9. Unseen-case demonstrations

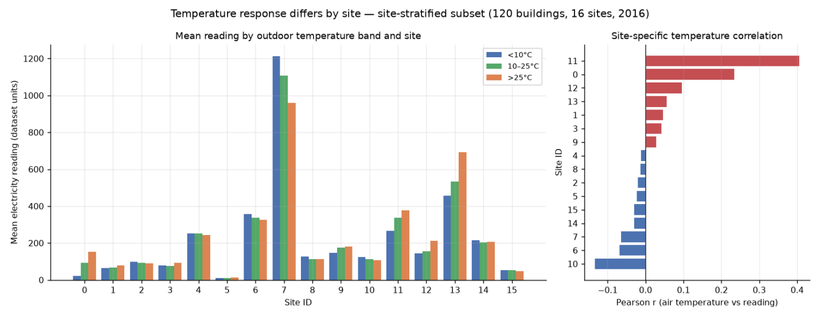
Case	Actual	Predicted	Error
Building 1088, 19 Dec 15:00	3,427.60	3,417.33	0.30%
Building 386, 9 Dec 06:00	41.16	41.16036	0.0009%

The near-exact second case is not proof of generalization; trees can match a leaf closely. The building-holdout experiment is the real test.

10. Five engineering insights

- **Demand is right-skewed and building-dominated.** Median 59.05 vs mean 188.78; floor-area log correlation 0.750. Benchmark within use/size groups.
- **Occupancy schedules create a weekday/midday cycle.** Weekday 195.44 vs weekend 172.24; peak 13:00, trough 03:00. Target setback studies.
- **Temperature response is site-specific.** Pooled $r = -0.074$, but site r ranges −0.134 to +0.406. Use degree hours or site interactions.
- **Primary use and site dominate.** Median reading 3.75–157.20 by use; site means 10.24–1,156.11. Segment reporting.
- **Seen-building accuracy does not transfer.** R^2 0.970 seen vs −0.555 unseen. Recalibrate before new-building use.

Temperature response is site-specific



Pooled correlation masks opposite heating and cooling regimes.

11. Dashboard and reproducibility

Tableau-ready extracts: 1,030,967 electricity rows; 349,232 prediction rows; 144 MB total. Spec covers overview, demand, weather, model performance, and data quality.

Streamlit supplement: four pages (Overview, Weather & Buildings, Model Performance, Predict), smoke-tested with no exceptions.

Tableau workbook is pending user authoring. Notebook, tests, README, and technical report are in the repository.

12. Reproducibility and status

- 22 automated tests pass; the 23-cell notebook (13 code cells) executed with saved outputs.
- Commands: `python -m src.data_loading`; `python -m src.train_models`; `streamlit run dashboard/streamlit_app.py`.
- Full-data Colab path is documented; local subset results are the only metrics reported.
- Dataset approval and the Tableau workbook remain pending.

Conclusion. The model is a useful **seen-building monitoring and interpolation tool**, not a validated new-building predictor. Results are from a 120-building subset, one year, a winter test month, site-level weather, and substantially missing metadata.

Recommendations. Add past-only history features; add degree hours and site interactions; model zero/non-zero separately; calibrate residuals by building and load band; validate on a future year and a building holdout; build and verify the Tableau workbook.

Limitations. One year; one winter test month; site-level weather; missing metadata; dataset units not independently verified; zero inflation; no causal claims; no full-data metrics; no Tableau workbook yet.

References. ASHRAE (2019), *Great Energy Predictor III*, Kaggle. Miller et al. (2020), *Science and Technology for the Built Environment*, 26(10), 1427–1447. Amasyali & El-Gohary (2018), *Renewable and Sustainable Energy Reviews*, 81, 1192–1205.

Status: local subset complete; Tableau workbook pending; full-data run pending; dataset approval pending. Code and report: README.md, reports/technical_report.md, reports/engineering_insights.md.